Advanced Discrete Event Simulation in R

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SMDM 2026 (Oslo, Norway)

Disclosures

- We have no financial or other conflicts to declare
- Supported by the NCI CISNET Incubator and CISNET programs:
  - All by U01 CA265750-02 (CISNET bladder cancer)
  - Alarid-Escudero also by U01 CA265729 (CISNET gastric cancer); and U01 CA253913 (colorectal cancer)
- Trikalinos and Sereda developed and maintain the **nhppp** R package

Sunday 28th of June, 8:35 to 12:00

Time	Description	Discussant
[15 min]	(0) Introductions and administrivia	Trikalinos
[25 min]	(1) DES as a composition of point processes	Alarid-Escudero
[30 min]	(2) NHPPPs – key properties	Trikalinos
[30 min]	(3) Sampling from NHPPPs	Trikalinos
[15 min]	**Break**	
[80 min]	(4) Working with code -- Base `R` vs `nhppp`: Simulate first vs all events in $(a, b]$ -- Packaging a simple cancer DES -- The vectorized case	[All] Trikalinos Sereda Sereda
[5 min]	(5) Advanced Topic Teaser on self-excitatory processes: point processes that are not NHPPPs and when you may need them	Trikalinos
[15 min]	General Q & A	All

Administrivia

- Professional conduct
  - https://smdm.org/hub/page/smdm-conduct-policy
- Bathroom locations
- WiFi (strip the quotation marks; keep whitespace in password)
  - SSID: “conferences”
  - Password: “Leonhard Euler”
- Format of the course

For the hands-on part,

- You need R, preferably with an IDE such as R Studio.
- Install packages `data.table` and `nhppp (>=1.0.0)`.
  To install them from CRAN,

```
> install.packages("data.table")
> install.packages("nhppp")
```

- All materials are available at
  - Through this Dropbox link (full link in the email) through 28/07/2026. Password "**smdm_oslo**",
  - `https://github.com/ttrikalin/des-R-course`

  (`smdm_2026` release)

Learning objectives

Be able to discuss:

- How a basic DES is organized
- Three properties of NHPPPs (memorylessness, composability, and transmutation by transforming the time axis) that are important for simulation
- Sampling algorithms and their use via `R`’s **nhppp** package